

Project Planning – Sprint / Story Point / Velocity Logic

Nutrition Assistant

Sprint Planning

A Sprint is a fixed period or duration in which a team works to complete a set of tasks.

An Epic is a big task or project that is too large to complete in one sprint. It is broken down into smaller tasks (stories) that can be completed over multiple sprints.

A Story is a small task. It is part of an Epic.

A Story Point is a number that represents how much effort a story takes to complete (usually in the form of a Fibonacci series).

1 – Very Easy Task

2 – Normal Task

3 – Moderate Task

5 – Difficult Task

Story Points used: 1, 2, 3, 5 (Fibonacci Series)

Sprint 1 – Project Setup & User Authentication

Project Setup (Epic 1)

Setting up React (Vite) frontend project (USN1) → 2

Setting up Node.js & Express backend project (USN2) → 2

Configuring MongoDB Atlas connection (USN3) → 2

User Authentication (Epic 2)

Building User Registration API (USN4) → 3

Building User Login API with JWT Authentication (USN5) → 5

Total Story Points in Sprint 1

$= 2 + 2 + 2 + 3 + 5$

$= 14$ Story Points

Sprint 2 – Food Search & Nutrition Tracking

Food Search (Epic 3)

Building Food Search API (USN6) → 5

Displaying Nutritional Information (USN7) → 5

Nutrition Tracking (Epic 4)

Developing Calorie Tracking Module (USN8) → 5

Developing Macronutrient Analysis Module (USN9) → 3

Building User Profile Management (USN10) → 2

Total Story Points in Sprint 2

$= 5 + 5 + 5 + 3 + 2$

$= 20$ Story Points

Sprint 3 – Dashboard, Testing & Deployment

Dashboard Development (Epic 5)

Building Nutrition Dashboard (USN11) → 3

Displaying Daily Nutrition Summary (USN12) → 5

Application Optimization (Epic 6)

Database Optimization using MongoDB (USN13) → 4

Functional & Integration Testing (USN14) → 5

Cloud Deployment (Vercel & Render) (USN15) → 3

Total Story Points in Sprint 3

$= 3 + 5 + 4 + 5 + 3$

$= 20$ Story Points

Total Story Points

Sprint 1 = 14

Sprint 2 = 20

Sprint 3 = 20

Velocity

Velocity = Total Story Points Completed / Number of Sprints

Total Story Points

$= 14 + 20 + 20$

$= 54$

Number of Sprints

$= 3$

Velocity

$= 54 / 3$

$= 18$ Story Points per Sprint

Final Result

Sprint 1 = 14 Story Points

Sprint 2 = 20 Story Points

Sprint 3 = 20 Story Points

Total Story Points = 54

Team Velocity = 18 Story Points per Sprint